

Data Management 2025/2026

Project Proposal

Group Members

Davide De Blasio (2082600)
Enrico Battistoni (2272055)

Project Type

Data Warehousing (DW) — Multi-source ETL + DFM Model + Star Schema + OLAP Analysis

Datasets

Source	Description	Link
IMDb Public Datasets	Title basics, ratings, crew, cast (TSV format, updated weekly)	https://datasets.imdbws.com
The Movies Dataset (Kaggle)	Budget, revenue, production companies, languages, countries, keywords	https://www.kaggle.com/datasets/rounakbanik/the-movies-dataset

Description of Intended Work

We intend to design and implement a Data Warehouse for analyzing the film industry, integrating two heterogeneous data sources: IMDb Public Datasets and The Movies Dataset from Kaggle. The two sources will be reconciled through an ETL pipeline (Python scripts + SQL) that handles ID mapping between IMDb and TMDb identifiers, data cleaning, genre normalization, and NULL management, producing a unified Reconciled Layer in PostgreSQL. Starting from this layer, we will design the Dimensional Fact Model (DFM) with at least four non-trivial dimensions: Time (Day → Month → Quarter → Year → Decade), Genre (Genre → GenreGroup), Production (Film Company → Country → Continent), and Director (Director → Nationality → Region), each featuring structured hierarchies suitable for OLAP drill-down and roll-up operations. We will then implement the Star Schema in PostgreSQL, motivating the design choice, and populate it via SQL ETL queries from the reconciled layer. Finally, we will conduct OLAP sessions to extract non-trivial insights such as revenue trends by genre over decades, ROI patterns across production company sizes, and the relationship between production budget and audience rating.